

The Robust H_∞ Linear Fault-Tolerant Recursive Filter Design of Quantum Communication System

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The robust H_∞ linear recursive filter design strategy in (22) is that the effect of general uncertainty, residual and noise $\tilde{\varepsilon}(k)$ on the filtering error should be minimized by the design of filter gain $\bar{L}$ of the linear recursive filter in (20). The initial Lyapunov energy function $\tilde{x}^T(0)P\tilde{x}(0)$ in the numerator in (22) means the robust H_∞ fault-tolerant recursive filter in (20) cannot affect the initial condition $\tilde{x}(0)$ and therefore the initial power $\tilde{x}^T(0)P\tilde{x}(0)$ should be extracted from the filtering error energy.

Consequently, the robust H_∞ fault-tolerant recursive filter design strategy in (22) is equivalent to determining the filter gain $\bar{L}$ in (20) such that the following filtering inequality is satisfied [16].

$$E \sum_{k=0}^{k_f} \{\tilde{x}^T(k)Q\tilde{x}(k) - \gamma^2 \tilde{\varepsilon}^T(k)\tilde{\varepsilon}(k)\} \leq \tilde{x}^T(0)P\tilde{x}(0) \quad (23)$$

Therefore, the design procedure of optimal H_∞ robust filter in (20) for the equivalent linear augmented discrete-time stochastic quantum communication system (18) can be divided into the following 3 steps: (i) The first step is to specify L to solve the following minimax Nash game problem for the filtering error system in (21).

$$J = E \sum_{k=0}^{k_f} \{\tilde{x}^T(k)Q\tilde{x}(k) - \gamma^2 \tilde{\varepsilon}^T(k)\tilde{\varepsilon}(k)\} \quad (24)$$

(ii) The second step is to guarantee an upper bound on the H_∞ filtering error energy J by the initial error energy power, i.e. $J \leq \tilde{x}^T(0)P\tilde{x}(0)$ in (33). (iii) The third step is to minimize the upper bound γ^2 in (22) as small as possible to achieve the optimal H_∞ robust fault-tolerant linear recursive filter of quantum discrete-time networked system.

Based on the above three steps procedure of robust linear fault-tolerant recursive filter design, we get the following result.

Theorem 1: For the linear recursive filter in (20), (i) if the filter gain $\bar{L}$ can be determined such that the following BMI holds

$$\begin{bmatrix} Q - P + (\bar{A} - \bar{L}\bar{C})^T P (\bar{A} - \bar{L}\bar{C}) & (\bar{A} - \bar{L}\bar{C})^T P \\ P(\bar{A} - \bar{L}\bar{C}) & -\gamma^2 I + P \end{bmatrix} \leq 0 \quad (25)$$

then the robust H_∞ fault-tolerant filtering objective in (22) is achieved. (ii) if the general residual $\tilde{\varepsilon}(k)$ in (21) is of finite energy, i.e., $\tilde{\varepsilon}(k) \in l_2[0, \infty)$ then $\tilde{x}(k) \rightarrow 0$ in m.s.s. as $k \rightarrow \infty$.

Based on Theorem 1, we need to find the solution of $\bar{L}$ and $P = P^T > 0$ from the BMI in (25). Before further derivation, the following lemma is required.

Lemma 2 (Schur Complement [16]): For matrices, $A = A^T$ and $C = C^T$ with adequate dimensions, and if C^{-1} exists, the LMI $\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \geq 0$, is equivalent to $C > 0$ and $A - BC^{-1}B^T \geq 0$.

The main result H_∞ linear fault-tolerant recursive filter design for quantum communication.

Based on Lemma 2 and Theorem 1, the robust optimal H_∞ fault-tolerant linear recursive filter design of quantum discrete-time communication system in Fig. 2 with the minimum attenuation level γ^2 obtained in Step (iii) is given as the following BMI-constrained optimization problem.

$$\gamma_0^2 = \min_{P>0, \bar{L}} \gamma^2 \quad (26)$$

subject to BMI in (25)

Since the BMI in (25) contains nonlinear unknown term $\bar{L}^T P \bar{L}$ and $P \bar{L}$, there exist several local minima. The solution of BMI-constrained optimization problem in (26) is conventional BMI solving algorithms are easily attracted by a local minimum. To overcome the local minimum problem of (26), let us denote $\bar{Z} = P \bar{L}$. Then we get the following main result of the optimal robust H_∞ fault-tolerant linear recursive filter design of the discrete-time quantum system under multiple weak measurements multiple channel interference and time-varying delay in Fig. 2.

Theorem 2: (i) The optimal robust H_∞ fault-tolerant linear recursive filter design problem in (20) and (22) of the quantum discrete-time communication system under multiple weak measurements and time-varying delay in (8) is solved by the following LMI-constrained optimization problem.

$$\gamma_0^2 = \min_{P>0, Z} \gamma^2 \quad (27)$$

s.t.

$$\begin{bmatrix} Q - P & (P\bar{A} - \bar{Z}\bar{C})^T & (P\bar{A} - \bar{Z}\bar{C})^T \\ P\bar{A} - \bar{Z}\bar{C} & -P & 0 \\ P\bar{A} - \bar{Z}\bar{C} & 0 & -\gamma^2 I + P \end{bmatrix} \leq 0 \quad (28)$$

(ii) If the optimal solutions $P^* > 0$ and Z^* are obtained from the LMI-constrained optimization problem in (27) and (28), then the optimal filter gain is recovered as $\bar{L}^* = P^{*-1}Z^*$ which yields the optimal robust H_∞ fault-tolerant linear recursive filter in (20).

According to the quantum communication system in Fig. 2. There are two robust linear quantum filters to be designed: one is at the quantum transmitter to estimate quantum state $\hat{x}(k)$ for the optimal quadratic reference tracking control

law $u_m(k)$ in remark 1 to let the state of the transmitted quantum particle optimally track the reference signal $x_r(k)$ to be transmitted. Another at the receiver to estimate the state of received quantum photons or electron via (i.e., to reconstruct the quantum information at receiver).

Since the system models in (7) of quantum communication system with multiple continuous weak measurement (8) at the transmitter and receiver are similar except the time-varying delay τ_k and channel interference $f(k)$, i.e., $b(k)$ in (11) has some difference when multiple continuous weak measurements at transmitter or at receiver. Actually, from the linear regression model in (13) or the state space in (15), only the modeling residual $\varepsilon_b(k)$ has some difference as mentioned in remark 1. Therefore, the augmented discrete-time quantum communication system in (18) is almost the same for two linear recursive filters in the transmitter and receiver except the modeling residual vector $\varepsilon(k)$. Since the proposed robust H_∞ linear fault-tolerant recursive filter design strategy in (22) holds for all $\bar{\varepsilon}(k) = [\varepsilon^T(k)(Ln(k))^T]^T \in l_2[0, k_f]$, the linear recursive filter in (20) based on the LMIs-constrained optimization problem can be also employed in the quantum transmitter to achieve the robust H_∞ state estimation of the quantum particle for the optimal reference quantum tracking control of quantum particle to be transmitted.

Simulation results will be presented to confirm the proposed quantum communication system design based on robust H_∞ observer-based reference tracking control at transmitter and robust H_∞ linear recursive filter at receiver in the presentation of CACS 2026.